



DO-IT-YOURSELF: CIRCUIT IDEA

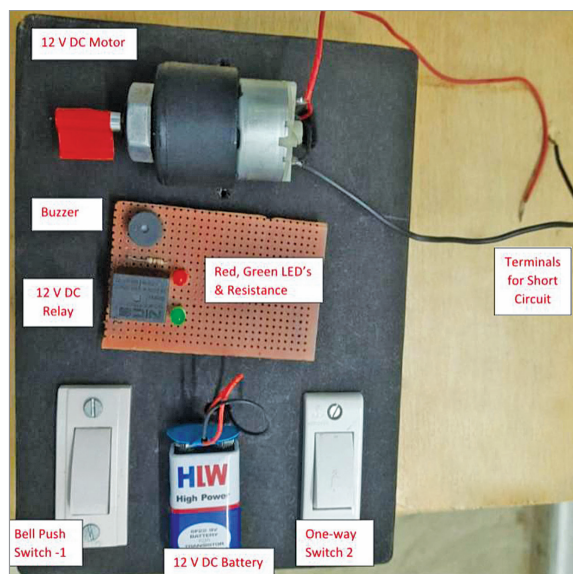


Fig. 3: Author's prototype of protection circuit

will glow, and the buzzer will sound, indicating that the battery is charged.

Step 2. Press the momentary push-to-on switch S2 while ensuring that S1 is closed. This action

connects the negative supply to terminal (A1) of the relay coil, energising relay RL1. The common terminal of RL1 disconnects from the normally closed (NC) contact and connects to the normally open (NO) contact. In this state, the green LED (LED1) will glow, and the motor will turn on, starting the vehicle's operation.

Step 3. In this step, manually short-circuit the motor terminals to simulate a short circuit condition in an EV. The

voltage across the motor will drop to zero, causing relay RL1 to de-energise, as depicted in Fig. 2.

Step 4. Consequently, due to the short circuit, relay RL1 is

discharged, and both the red LED (LED1) and the piezo buzzer are enabled. The motor is disconnected from the battery, ensuring the protection of both the motor and the battery.

This circuit effectively resolves the problem of short circuits in electrical vehicles. By using this method, you can save the motor from damage caused by short circuits and prevent battery discharge. This, in turn, extends the lifespan of batteries, which are a crucial component of EVs. Considering the high cost of batteries, this circuit offers valuable protection for both the motor and the batteries of electrical vehicles. The author's prototype of the circuit protection for the electrical motor and battery in an electric vehicle is shown in Fig. 3. **EFY**



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